Chef Classification from Recipe Text using DistilBERT

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October 6, 2025

1 Overview

We fine-tuned DistilBERT-base-uncased to assign each recipe to one of six chefs. The original 2,999-row dataset contained duplicated texts; removing 14 exact matches before splitting left 2,985 unique samples. We stratified an 80/20 train-validation split and evaluated with accuracy and macro-F1. The best checkpoint—captured by early stopping after roughly five epochs—achieves 92.13% validation accuracy and 0.916 macro-F1, exceeding the strongest TF-IDF baseline by about 49 percentage points.

2 Data Preparation

Our preprocessing focuses on consistency and leakage prevention. Recipe fields stored as lists (ingredients, steps, tags) are parsed and trimmed. We concatenate five text segments in the order *name* $\rightarrow$ *ingredients* $\rightarrow$ *tags* $\rightarrow$ *description* $\rightarrow$ *steps*; shorter sections lead, ensuring that the steps (which average 46% of tokens) are truncated last when a recipe exceeds the 512-token window. After deduplication, class counts range from 365 to 801 recipes (Appendix Figure 4). Token statistics remain manageable (median 234 tokens, 95th percentile 418), so the DistilBERT tokenizer rarely truncates content.

3 Model and Optimisation Choices

We retained DistilBERT's default classification head: dropout 0.2, a $768 \rightarrow 768$ projection with ReLU, and a linear layer mapping to six logits (Figure 1). The optimiser is AdamW with learning rate 2×10^{-5}, weight decay 0.01, linear warm-up over 6% of steps, and gradient clipping at 1.0. To keep the Mac training run stable we use "chill mode" batch sizes of 8/16 (train/eval) with dynamic padding. Evaluation occurs every 100 steps; if macro-F1 fails to improve twice in a row we stop, keeping the best checkpoint. This tighter cadence proved more informative than fixed-epoch schedules—later epochs fluctuate but do not improve the macro-F1 curve (Appendix Figure 3).

4 Evaluation

Table 1: Validation accuracy and macro-F1.

Method	**Accuracy**	**Macro-F1**
TF-IDF (description)	30.0%	—
TF-IDF (all fields)	43.0%	—
DistilBERT (ours)	**92.13%**	**0.916**

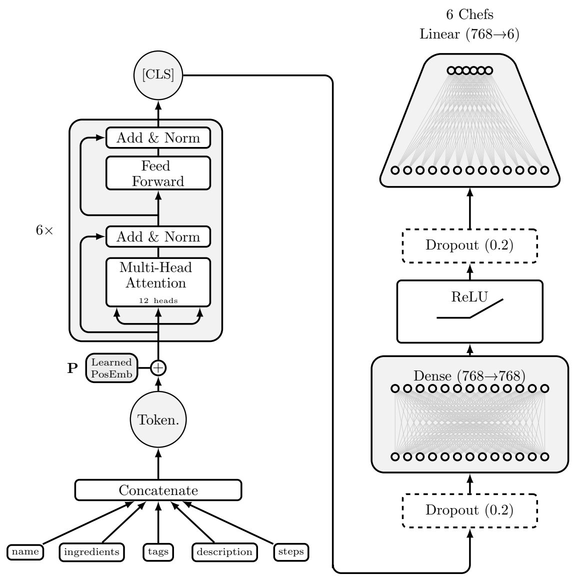

Figure 1: Text-only DistilBERT pipeline used for chef classification.

Validation predictions remain well-balanced: class-wise precision and recall exceed 0.85, and the blind test distribution deviates by at most 2.3 percentage points from the training prior (Appendix Figure 5). Chef 1533 is modestly over-predicted, while Chef 3288 is slightly under-predicted—both chefs specialize in quick appetisers and make-ahead dishes, so confusion is expected. Figure 3 (appendix) shows that accuracy and macro-F1 improve steadily through epoch 4; thereafter the curves stabilise, and later checkpoints bring no gains.

5 Key Takeaways and Next Steps

- **Deduplication mattered.** Earlier experiments that split before deduplicating produced optimistic metrics. Removing the 14 overlapping recipes eliminated this leakage.

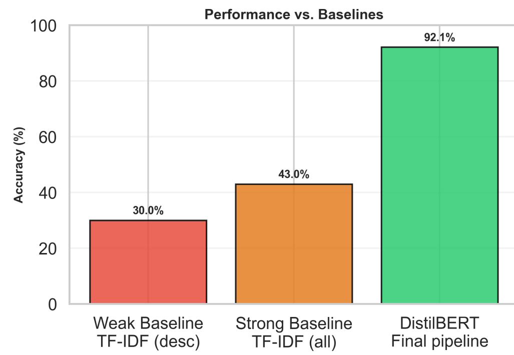

Figure 2: Accuracy improvement over TF-IDF baselines.

- **Simple head, careful schedule.** Returning to the default ReLU head and monitoring macro-F1 every 100 steps was enough to reach 92%; more complex heads were unnecessary on this dataset size.

- **Future improvements.** Structured cues (ingredient quantities, durations) and cross-validation could further stabilise scores. Interpretability analysis—e.g., attention heat maps—would help confirm the model looks beyond obvious keywords such as "diabetic cooking" or "OAMC".

Bibliography

References

Appendix: Supporting Figures

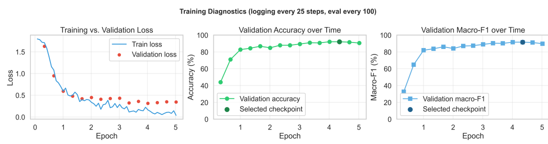

Figure 3: Training diagnostics: loss, validation accuracy, and macro-F1 at each evaluation (every 100 steps).

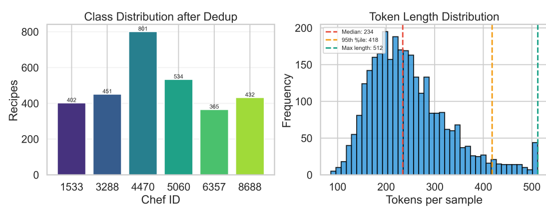

Figure 4: Dataset overview: deduplicated class counts (left) and token length distribution (right).

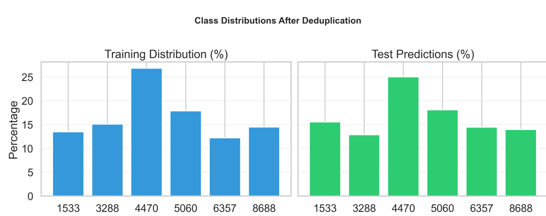

Figure 5: Comparison between training distribution (left) and blind test predictions (right).